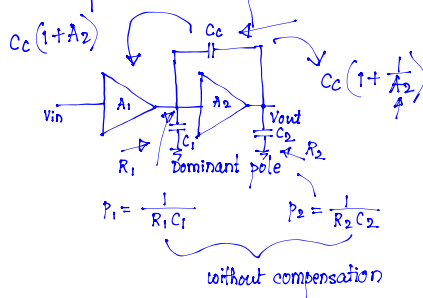
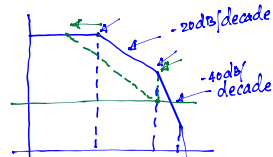
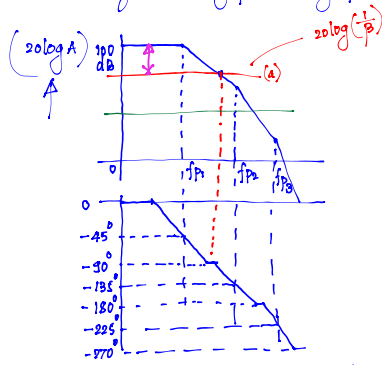


Frequency compensation :-

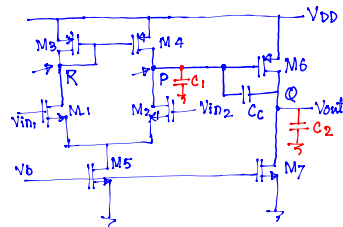
$$20 \log A - 20 \log \frac{1}{p} = 20 \log A p$$



After compensation

$$p_1' = -\frac{1}{R_1 [C_1 + (1+A_2)C_c]}$$

$$\approx -\frac{1}{R_1 A_2 C_c}$$



Without compensation

$$p_1 = -\frac{1}{(r_{o2} || r_{o4}) C_1}$$

$$p_2 = -\frac{1}{(r_{o6} || r_{o7}) C_2}$$

$$\frac{V_{out}(s)}{V_{in}} = \frac{g_{m1} g_{m2} R_1 R_2 (1 - s \frac{C_c}{g_{m2}})}{1 + s [C_1 + (1 + g_{m2} R_2) C_c] R_1 + R_2 (C_2 + C_c) + s^2 R_1 R_2 [C_1 C_2 + C_2 C_c + C_1 C_c]}$$

$$D = \left(1 + \frac{s}{\omega_{p1}}\right) \left(1 + \frac{s}{\omega_{p2}}\right)$$

$$= 1 + \left(\frac{1}{\omega_{p1}} + \frac{1}{\omega_{p2}}\right) s + \frac{s^2}{\omega_{p1} \omega_{p2}}$$